A Journal Recommendation System for Computer Science Articles Using Text Mining and Clustering Techniques

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Abstract —Journal recommendation, text mining, TF-IDF, classification, clustering.

This project presents a journal recommendation system designed for computer science research articles. The main objective is to assist researchers in identifying the most suitable journals for their work based on the abstract of their articles. A dataset consisting of 7711 articles from 175 journals was utilized.

The methodology involves preprocessing the text data by removing HTML tags and converting text into lowercase. Then, TF-IDF vectorization was applied to transform textual data into numerical features. Machine learning techniques, including clustering and classification models such as Multinomial Naive Bayes, were used to analyze patterns in the data.

Additionally, topic clustering was performed using K-Means to group similar research areas. Based on the trained model, when a user inputs an abstract, the system predicts and returns the top five most relevant journals.

The results demonstrate that the system is capable of identifying meaningful patterns in research topics, although performance is affected by class imbalance in the dataset. This project highlights the importance of text mining techniques in solving real-world research problems.

I. Introduction

Selecting the appropriate journal for publishing a research article is a critical decision for researchers. The choice of journal affects the visibility, impact, and dissemination of the research work. However, due to the large number of available journals and the diversity of research topics, identifying the most suitable journal has become a challenging task.

With the rapid growth of scientific publications, traditional manual methods for selecting journals are no longer efficient. Researchers often rely on experience or limited knowledge, which may lead to suboptimal decisions. Therefore, there is a need for intelligent systems that can assist researchers in making data-driven decisions.

In this project, a journal recommendation system is developed for computer science articles. The system analyzes the abstract of a research paper and suggests the most relevant journals based on text mining and machine learning techniques.

The proposed approach utilizes TF-IDF for feature extraction, classification models for prediction, and clustering methods to identify topic groups. By combining these techniques, the system aims to improve the accuracy and efficiency of journal selection.

II. Literature Review

Several studies have focused on the problem of journal recommendation and text classification using machine learning techniques. With the increasing number of scientific publications, researchers have explored different approaches to analyze textual data and identify relevant research areas.

Previous work has shown that text mining techniques such as TF-IDF are effective in extracting meaningful features from research articles. These features are commonly used in classification models such as Naive Bayes, Support Vector Machines, and Logistic Regression to predict categories or labels.

In addition, clustering methods such as K-Means have been widely used to group similar documents based on their content. This helps in identifying hidden topic structures within large datasets. Many studies have demonstrated that clustering can improve the understanding of research domains and assist in recommendation systems.

Some research has specifically addressed journal recommendation systems by combining classification and similarity-based approaches. These systems analyze article abstracts and match them with previously published works to suggest suitable journals.

Despite these advancements, challenges such as class imbalance, large number of categories, and overlapping topics still affect the performance of these systems. In this project, both classification and clustering techniques are used to address these challenges and improve journal recommendation accuracy.

III. Methodology

In this project, a journal recommendation system was developed using text mining and machine learning techniques. The methodology consists of several main steps including data preprocessing, feature extraction, classification, and clustering.

A. Data Preprocessing

The dataset consists of 7711 academic articles collected from 175 computer science journals. Each record contains the abstract text and the corresponding journal name.

The dataset contains multiple entities such as article abstracts, journal names, keywords, and subject categories. In this project, the focus was primarily on the abstract text (AcademicRecordAbstract) and journal names (Publication), as these provide the most direct information for journal recommendation. Other entities such as keywords and subject categories were not used in this implementation but could be incorporated in future work to improve the system performance.

Initially, the dataset was loaded using Python and processed using the pandas library. Since the abstract texts contained HTML tags, a cleaning step was applied to remove these tags. In addition, all text was converted to lowercase to ensure consistency during analysis.Initially, the dataset was loaded using Python and processed using the pandas library. Since the abstract texts contained HTML tags, a cleaning step was applied to remove these tags. In addition, all text was converted to lowercase to ensure consistency during analysis.

B. Feature Extraction

To convert textual data into numerical form, the TF-IDF (Term Frequency–Inverse Document Frequency) technique was used. This method assigns weights to words based on their importance in the document and across the dataset.

A TF-IDF vectorizer was applied with a maximum of 5000 features and a combination of unigrams and bigrams (ngram_range = (1,2)) to capture more contextual information.

C. Data Splitting

The dataset was divided into training and testing sets using an 80/20 split. Only journals with at least two articles were included to avoid errors during stratified sampling.

D. Classification Model

A Multinomial Naive Bayes classifier was used to predict the journal based on the abstract text. This model is commonly used in text classification problems due to its efficiency and performance.

The model was trained using the TF-IDF features extracted from the training data, and predictions were generated for the test data.

E. Clustering

In addition to classification, K-Means clustering was applied to group similar research topics. The dataset was divided into 10 clusters based on content similarity.

Each cluster was analyzed by extracting the top keywords, which represent the main topic of that group. This helps in understanding the distribution of research areas within the dataset.

F. Journal Recommendation

Based on the trained model, when a new abstract is provided, the system processes the text, transforms it using TF-IDF, and predicts the most relevant journals. The top 5 journals are then ranked and presented to the user.

IV. Results

The performance of the proposed journal recommendation system was evaluated using classification metrics. The Multinomial Naive Bayes model achieved an accuracy of approximately 0.25 on the test dataset.

Although the accuracy appears relatively low, this result is expected due to the large number of journal classes and the imbalance in the dataset. Many journals have a limited number of articles, which makes it difficult for the model to learn strong patterns for all categories.

The classification report shows that some journals achieved better precision and recall values compared to others, especially those with a higher number of training samples. In contrast, journals with fewer samples often resulted in zero precision and recall, indicating that the model could not correctly predict them.

In addition to classification, clustering results provided meaningful insights into the dataset. Using K-Means clustering, the abstracts were grouped into 10 clusters representing different research topics. The top keywords for each cluster revealed distinct subject areas such as:

- Cloud computing and security
- Wireless communication and networks
- Data mining and big data
- Fuzzy decision-making systems
- Optimization algorithms
- Machine learning and classification
- Image processing and computer vision These clusters

These clusters demonstrate that the model successfully identified underlying research themes within the dataset.

Overall, the system is capable of capturing general patterns in research topics and providing reasonable journal recommendations, especially for well-represented categories.

V. Conclusion

In this project, a journal recommendation system for computer science articles was successfully developed using text mining and machine learning techniques. The system processes article abstracts and predicts the most relevant journals based on learned patterns from the dataset.

TF-IDF was used for feature extraction, while Multinomial Naive Bayes was applied as the classification model. Additionally, K-Means clustering was used to identify underlying research topics within the dataset.

The results showed that the system can capture general trends in research areas and provide useful journal recommendations. However, the performance was limited due to class imbalance and the large number of journal categories.

Future improvements may include using more advanced machine learning models, such as deep learning approaches, improving data balance, and incorporating additional entities such as keywords and subject categories to enhance recommendation accuracy.

Overall, this project demonstrates the effectiveness of text mining techniques in solving real-world research problems and highlights the potential for further development in intelligent recommendation systems.

IV. Results

[29]:

	Rank	JournalName	Score
0	1	INTERNATIONAL JOURNAL OF PATTERN RECOGNITION A...	0.004505
1	2	JOURNAL OF INTELLIGENT INFORMATION SYSTEMS	0.004450
2	3	DISTRIBUTED COMPUTING	0.004286
3	4	JOURNAL OF COMPUTER AND SYSTEM SCIENCES	0.004248
4	5	COMPUTER METHODS AND PROGRAMS IN BIOMEDICINE	0.004204

VI. References

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